

Unipotent generators at parameter zero are trivial in the group $G(\mathbf{m})$: a machine-checked proof

Abstract

We present a formal verification of Lemma 3.1 from the source text on the group $G(\mathbf{m})$. This group is defined by generators and relations that encode copies of GL_2 indexed by the coefficients $c(j)$ of the modular function $J(q)$. The lemma states that the unipotent generators $X_{-1}(0)$, $Y_{-1}(0)$, $X_{\ell,jk}(0)$ and $Y_{\ell,jk}(0)$ are trivial. The formalization builds the full presentation as a presented group over the free group on the generating symbols. It takes the coefficient function as an arbitrary parameter $c: \mathbb{N} \rightarrow \mathbb{N}$ and needs no hypothesis on c . It proves all four parts, using only the additivity relations (Re:1a), (Re:1b), (Im:1a) and (Im:1b).

Source	Carbone et al., G-Simple (draft), GSimple_Lemma_3_1
Lean	leanprover/lean4:v4.23.0, Mathlib 37df177aaa
Machine checks	compiles; no sorry; statement matches target; module build
Axioms	propext, Classical.choice, Quot.sound
Independent review	FAITHFUL (claude-opus-5-5)
Agents	captain claude-sonnet-5-5; solvers aiprover; report claude-opus-5-5
Run	srv_20261004_123233_GSimple_Lemma_3_1

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1 Statement

Let $\mathbb{C}$ denote the complex numbers and $\mathbb{N}$ the natural numbers. Let $c(j)$ denote the coefficient of q^j in the modular function

$$J(q) = \sum_{j \geq -1} c(j)q^j = \frac{1}{q} + 196884q + 21493760q^2 + 864299970q^3 + \dots,$$

so that $c(-1) = 1$, $c(0) = 0$, $c(1) = 196884$, and so on. Define the index set

$$E = \{(\ell, j, k) \mid j \in \mathbb{N}, 1 \leq k \leq c(j), 0 \leq \ell < j\}$$

and, for $(\ell, j, k) \in E$, the complex number

$$c_{\ell j} := (-1)^{\ell+1} \binom{j-1}{\ell} (\ell+1)(j-\ell).$$

Let $\mathcal{X}$ be the set of symbols

$$\mathcal{X} := \{H_1(s), H_2(s), X_{-1}(u), Y_{-1}(u), X_{\ell,jk}(u), Y_{\ell,jk}(u) \mid s \in \mathbb{C}^\times, u \in \mathbb{C}, (\ell, j, k) \in E\},$$

and let $F(\mathcal{X})$ be the free group on $\mathcal{X}$. The symbols $X_{-1}(u)$, $Y_{-1}(u)$, $X_{\ell,jk}(u)$, $Y_{\ell,jk}(u)$ are the *unipotent generators*. The commutator is $(a, b) := aba^{-1}b^{-1}$.

The set $\mathcal{R} \subseteq F(\mathcal{X})$ and the auxiliary elements $\tilde{w}_{-1}(s)$ and $\tilde{w}_{\ell,jk}(s)$ are defined as follows. The relations hold for all $s, t \in \mathbb{C}^\times$, $u, v \in \mathbb{C}$ and $(\ell, j, k), (m, p, q) \in E$.

$$\tilde{w}_{-1}(s) := X_{-1}(s) Y_{-1}(-s^{-1}) X_{-1}(s), \quad \tilde{w}_{-1} := \tilde{w}_{-1}(1), \quad (\text{Re:0})$$

$$\tilde{w}_{\ell,jk}(s) := X_{\ell,jk}(s) Y_{\ell,jk}\left(\frac{-s^{-1}}{c_{\ell j}}\right) X_{\ell,jk}(s), \quad \tilde{w}_{\ell,jk} := \tilde{w}_{\ell,jk}(1), \quad (\text{Im:0})$$

Relations that define a subgroup $\text{GL}_2(-1)$:

$$H_1(s)H_1(t) = H_1(st), \quad (\text{H:1a})$$

$$H_2(s)H_2(t) = H_2(st), \quad (\text{H:1b})$$

$$H_1(s)H_2(t) = H_2(t)H_1(s), \quad (\text{H:2})$$

$$X_{-1}(u)X_{-1}(v) = X_{-1}(u+v), \quad (\text{Re:1a})$$

$$Y_{-1}(u)Y_{-1}(v) = Y_{-1}(u+v), \quad (\text{Re:1b})$$

$$Y_{-1}(-t)X_{-1}(s)Y_{-1}(t) = X_{-1}(-t^{-1})Y_{-1}(-t^2s)X_{-1}(t^{-1}), \quad (\text{Re:2})$$

$$\tilde{w}_{-1}(s)\tilde{w}_{-1} = H_1(-s)H_2(-s^{-1}), \quad (\text{Re:3})$$

$$\tilde{w}_{-1}X_{-1}(u)\tilde{w}_{-1}^{-1} = Y_{-1}(-u), \quad (\text{Re:4a})$$

$$\tilde{w}_{-1}Y_{-1}(u)\tilde{w}_{-1}^{-1} = X_{-1}(-u), \quad (\text{Re:4b})$$

$$\tilde{w}_{-1}H_1(s)\tilde{w}_{-1}^{-1} = H_2(s), \quad (\text{Re:5a})$$

$$\tilde{w}_{-1}H_2(s)\tilde{w}_{-1}^{-1} = H_1(s), \quad (\text{Re:5b})$$

$$H_1(s)X_{-1}(u)H_1(s)^{-1} = X_{-1}(su), \quad (\text{Re:6a})$$

$$H_2(s)X_{-1}(u)H_2(s)^{-1} = X_{-1}(s^{-1}u), \quad (\text{Re:6b})$$

$$H_1(s)Y_{-1}(u)H_1(s)^{-1} = Y_{-1}(s^{-1}u), \quad (\text{Re:6c})$$

$$H_2(s)Y_{-1}(u)H_2(s)^{-1} = Y_{-1}(su), \quad (\text{Re:6d})$$

Relations that define subgroups $\text{GL}_2(\ell, j, k)$:

$$X_{\ell,jk}(u)X_{\ell,jk}(v) = X_{\ell,jk}(u+v), \quad (\text{Im:1a})$$

$$Y_{\ell,jk}(u)Y_{\ell,jk}(v) = Y_{\ell,jk}(u+v), \quad (\text{Im:1b})$$

$$Y_{\ell,jk}(-t)X_{\ell,jk}(s)Y_{\ell,jk}(t) = X_{\ell,jk}\left(\frac{-t^{-1}}{c_{\ell j}}\right)Y_{\ell,jk}(-c_{\ell j}t^2s)X_{\ell,jk}\left(\frac{t^{-1}}{c_{\ell j}}\right), \quad (\text{Im:2})$$

$$\tilde{w}_{\ell,jk}(s^{(\ell+1)(j-\ell)})\tilde{w}_{\ell,jk} = H_1((-s)^{j-\ell})H_2((-s)^{\ell+1}), \quad (\text{Im:3})$$

$$\tilde{w}_{\ell,jk}X_{\ell,jk}(u)\tilde{w}_{\ell,jk}^{-1} = Y_{\ell,jk}\left(\frac{-u}{c_{\ell j}}\right), \quad (\text{Im:4a})$$

$$\tilde{w}_{\ell,jk}Y_{\ell,jk}(u)\tilde{w}_{\ell,jk}^{-1} = X_{\ell,jk}(-c_{\ell j}u), \quad (\text{Im:4b})$$

$$\tilde{w}_{\ell,jk}H_1(s^{j-\ell})\tilde{w}_{\ell,jk}^{-1} = H_2(s^{-(\ell+1)}), \quad (\text{Im:5a})$$

$$\tilde{w}_{\ell,jk}H_2(s^{\ell+1})\tilde{w}_{\ell,jk}^{-1} = H_1(s^{-(j-\ell)}), \quad (\text{Im:5b})$$

$$H_1(s)X_{\ell,jk}(u)H_1(s)^{-1} = X_{\ell,jk}(s^{\ell+1}u), \quad (\text{Im:6a})$$

$$H_2(s)X_{\ell,jk}(u)H_2(s)^{-1} = X_{\ell,jk}(s^{j-\ell}u), \quad (\text{Im:6b})$$

$$H_1(s)Y_{\ell,jk}(u)H_1(s)^{-1} = Y_{\ell,jk}(s^{-(\ell+1)}u), \quad (\text{Im:6c})$$

$$H_2(s)Y_{\ell,jk}(u)H_2(s)^{-1} = Y_{\ell,jk}(s^{-(j-\ell)}u), \quad (\text{Im:6d})$$

Relations between the unipotent generators:

$$(X_{-1}(u), X_{j-1,jk}(v)) = 1, \quad (\text{U:1a})$$

$$(Y_{-1}(u), Y_{j-1,jk}(v)) = 1, \quad (\text{U:1b})$$

$$(Y_{-1}(u), X_{0,jk}(v)) = 1, \quad (\text{U:1c})$$

$$(X_{-1}(u), Y_{0,jk}(v)) = 1, \quad (\text{U:1d})$$

$$(X_{\ell,jk}(u), Y_{m,pq}(v)) = 1 \quad \text{if } j \neq p, \text{ or } k \neq q, \text{ or } |\ell - m| > 1, \quad (\text{U:2})$$

$$(X_{1,2k}(u), Y_{0,2k}(v)) = X_{-1}(uv), \quad (\text{U:3a})$$

$$(X_{0,2k}(u), Y_{1,2k}(v)) = Y_{-1}(uv), \quad (\text{U:3b})$$

Action of $\tilde{w}_{-1}$ on the unipotent generators:

$$\tilde{w}_{-1}X_{\ell,jk}(u)\tilde{w}_{-1}^{-1} = X_{j-1-\ell,jk}((-1)^{j-\ell-1}u), \quad (\text{U:4a})$$

$$\tilde{w}_{-1}Y_{\ell,jk}(u)\tilde{w}_{-1}^{-1} = Y_{j-1-\ell,jk}((-1)^{j-\ell-1}u). \quad (\text{U:4b})$$

The group $G(\mathfrak{m})$ is the group given by this presentation:

$$G(\mathfrak{m}) = \langle \mathcal{X} \mid \mathcal{R} \rangle = F(\mathcal{X})/N_{\mathcal{R}},$$

where $N_{\mathcal{R}}$ is the normal closure of $\mathcal{R}$.

Lemma 1.1 (Lemma 3.1 of the source). *In $G(\mathfrak{m})$ we have:*

$$(a) \ X_{-1}(0) = 1;$$

$$(b) \ Y_{-1}(0) = 1;$$

$$(c) \ X_{\ell,jk}(0) = 1 \text{ for all } (\ell, j, k) \in E;$$

$$(d) \ Y_{\ell,jk}(0) = 1 \text{ for all } (\ell, j, k) \in E.$$

The source proves (a) by setting $u = v = 0$ in (Re:1a), which gives $X_{-1}(0)^{-1}X_{-1}(0)X_{-1}(0) = 1$, and hence $X_{-1}(0) = 1$. Parts (b), (c) and (d) follow in the same way from (Re:1b), (Im:1a) and (Im:1b).

2 Formalization

Throughout, $c: \mathbb{N} \rightarrow \mathbb{N}$ is an arbitrary function. Here $\mathbb{N} = \{0, 1, 2, \dots\}$. It plays the role of the coefficients $c(j)$ of J for $j \geq 0$. The value $c(-1)$ never enters the definition of E , so it is not represented. All objects below depend on c .

Definition 2.1 (Index set). E_c is the set of triples $(\ell, j, k) \in \mathbb{N}^3$ with $1 \leq k \leq c(j)$ and $\ell < j$.

Lean: IdxE

```
| def IdxE (c : ℕ → ℕ) : Type := {x : ℕ × ℕ × ℕ // 1 ≤ x.2.2 ∧ x.2.2 ≤ c x.2.1 ∧ x.1 < x.2.1}
```

The condition $\ell < j$ forces $j \geq 1$. Therefore it makes no difference whether the source's $\mathbb{N}$ contains 0.

Definition 2.2 (Generating symbols). $\mathcal{X}_c$ is the disjoint union of six families of symbols:

- $H_1(s)$ and $H_2(s)$ for $s \in \mathbb{C}^\times$;
- $X_{-1}(u)$ and $Y_{-1}(u)$ for $u \in \mathbb{C}$;
- $X_e(u)$ and $Y_e(u)$ for $e \in E_c$ and $u \in \mathbb{C}$.

For $e = (\ell, j, k)$ we also write $X_e(u) = X_{\ell, jk}(u)$ and $Y_e(u) = Y_{\ell, jk}(u)$.

Lean: Gen

```
| inductive Gen (c : ℕ → ℕ) : Type
| H1 : ℂ× → Gen c
| H2 : ℂ× → Gen c
| Xm : ℂ → Gen c
| Ym : ℂ → Gen c
| X : IdxE c → ℂ → Gen c
| Y : IdxE c → ℂ → Gen c
```

Definition 2.3 (The constants $c_{\ell j}$). For $\ell, j \in \mathbb{N}$,

$$c_{\ell j} = (-1)^{\ell+1} \binom{j-1}{\ell} (\ell+1)(j-\ell) \in \mathbb{C},$$

where $j-1$ and $j-\ell$ are computed as truncated subtraction in $\mathbb{N}$.

Lean: cc

```
| noncomputable def cc (l j : ℕ) : ℂ :=
  (-1 : ℂ) ^ (l + 1) * (Nat.choose (j - 1) l : ℂ) * (l + 1 : ℕ) : ℂ * (j - l : ℕ) : ℂ
```

For $(\ell, j, k) \in E_c$ we have $\ell < j$, so the truncated differences are the ordinary ones and $c_{\ell j}$ agrees with the source. Moreover $0 \leq \ell \leq j-1$ gives $\binom{j-1}{\ell} \geq 1$, so $c_{\ell j} \neq 0$. In particular the divisions by $c_{\ell j}$ below are genuine divisions.

Definition 2.4 (Index flip). $\phi: E_c \rightarrow E_c$ is the map $(\ell, j, k) \mapsto (j-1-\ell, j, k)$. Since $\ell < j$, the integer $j-1-\ell$ lies in $[0, j)$, so the image lies in E_c .

Lean: flipE

```
def flipE {c : ℕ → ℕ} (e : IdxE c) : IdxE c :=
  ⟨(e.1.2.1 - 1 - e.1.1, e.1.2.1, e.1.2.2), (e.2.1, e.2.2.1, by
    have h := e.2.2.2
    show e.1.2.1 - 1 - e.1.1 < e.1.2.1
    omega)⟩
```

The next six definitions are the images of the generators in the free group $F(\mathcal{X}_c)$. They are denoted by the same symbols.

Definition 2.5. $H_1(s) \in F(\mathcal{X}_c)$ for $s \in \mathbb{C}^\times$.

Lean: h1

```
def h1 (s : ℂ×) : FreeGroup (Gen c) := FreeGroup.of (Gen.H1 s)
```

Definition 2.6. $H_2(s) \in F(\mathcal{X}_c)$ for $s \in \mathbb{C}^\times$.

Lean: h2

```
def h2 (s : ℂ×) : FreeGroup (Gen c) := FreeGroup.of (Gen.H2 s)
```

Definition 2.7. $X_{-1}(u) \in F(\mathcal{X}_c)$ for $u \in \mathbb{C}$.

Lean: xm

```
def xm (u : ℂ) : FreeGroup (Gen c) := FreeGroup.of (Gen.Xm u)
```

Definition 2.8. $Y_{-1}(u) \in F(\mathcal{X}_c)$ for $u \in \mathbb{C}$.

Lean: ym

```
def ym (u : ℂ) : FreeGroup (Gen c) := FreeGroup.of (Gen.Ym u)
```

Definition 2.9. $X_e(u) \in F(\mathcal{X}_c)$ for $e \in E_c$ and $u \in \mathbb{C}$.

Lean: xx

```
def xx (e : IdxE c) (u : ℂ) : FreeGroup (Gen c) := FreeGroup.of (Gen.X e u)
```

Definition 2.10. $Y_e(u) \in F(\mathcal{X}_c)$ for $e \in E_c$ and $u \in \mathbb{C}$.

Lean: yy

```
| def yy (e : IdxE c) (u : ℂ) : FreeGroup (Gen c) := FreeGroup.of (Gen.Y e u)
```

Definition 2.11 ((Re:0)). For $s \in \mathbb{C}^\times$, $\tilde{w}_{-1}(s) = X_{-1}(s) Y_{-1}(-s^{-1}) X_{-1}(s) \in F(\mathcal{X}_c)$.

Lean: wm

```
| noncomputable def wm (s : ℂ×) : FreeGroup (Gen c) := xm (c := c) s * ym (-(s : ℂ)-1) * xm (c := c) s
```

Definition 2.12. $\tilde{w}_{-1} = \tilde{w}_{-1}(1)$.

Lean: wm1

```
| noncomputable def wm1 : FreeGroup (Gen c) := wm (c := c) 1
```

Definition 2.13 ((Im:0)). For $e = (\ell, j, k) \in E_c$ and $s \in \mathbb{C}^\times$,

$$\tilde{w}_e(s) = X_e(s) Y_e(-s^{-1}/c_{\ell j}) X_e(s).$$

Lean: ww

```
| noncomputable def ww (e : IdxE c) (s : ℂ×) : FreeGroup (Gen c) :=
  xx e s * yy e (-(s : ℂ)-1) / cc e.1.1 e.1.2.1 * xx e s
```

Definition 2.14. $\tilde{w}_e = \tilde{w}_e(1)$ for $e \in E_c$.

Lean: ww1

```
| noncomputable def ww1 (e : IdxE c) : FreeGroup (Gen c) := ww e 1
```

Definition 2.15 (Commutator). $(a, b) = aba^{-1}b^{-1}$ for $a, b \in F(\mathcal{X}_c)$.

Lean: comm'

```
| def comm' (a b : FreeGroup (Gen c)) : FreeGroup (Gen c) := a * b * a-1 * b-1
```

Definition 2.16 (Relator of an equation). For $a, b \in F(\mathcal{X}_c)$, the relator encoding $a = b$ is ab^{-1} .

Lean: eqr

```
| def eqr (a b : FreeGroup (Gen c)) : FreeGroup (Gen c) := a * b-1
```

Definition 2.17 (Conjugation). $a \cdot b = aba^{-1}$ for $a, b \in F(\mathcal{X}_c)$.

Lean: conj'

```
| def conj' (a b : FreeGroup (Gen c)) : FreeGroup (Gen c) := a * b * a-1
```

Definition 2.18 (The relators $\mathcal{R}_c$). $\mathcal{R}_c \subseteq F(\mathcal{X}_c)$ is defined as follows.

- Each equation $A = B$ among (H:1a)–(U:4b) contributes the relator AB^{-1} . This applies for all $s, t \in \mathbb{C}^\times$, $u, v \in \mathbb{C}$ and $e, f \in E_c$.
- Each relation $(a, b) = 1$ contributes (a, b) itself.

The relators are written exactly as displayed in [Lemma 1.1](#) and the relations preceding it, with $\tilde{w}_{-1}$ and $\tilde{w}_e$ as above. The following conventions apply.

- In (Im:5a), (Im:5b), (Im:6c) and (Im:6d), the negative powers s^{-n} are written as $(s^n)^{-1}$.
- In (Re:2) and (Im:2), the parameters s, t range over $\mathbb{C}^\times$.
- (U:1a) and (U:1b) are imposed for $e = (\ell, j, k)$ with $\ell + 1 = j$. (U:1c) and (U:1d) are imposed for e with $\ell = 0$.
- (U:2) is imposed for $e = (\ell, j, k)$ and $f = (m, p, q)$ whenever $j \neq p$, or $k \neq q$, or $|\ell - m| > 1$ (as integers).
- (U:3a) is imposed for $e = (1, 2, k)$ and $f = (0, 2, k)$. (U:3b) is imposed for $e = (0, 2, k)$ and $f = (1, 2, k)$.
- (U:4a) and (U:4b) use the flipped index $\phi(e)$ and the sign $(-1)^{j-\ell-1}$. Here $j - \ell - 1 \geq 0$ for $e \in E_c$.

Lean: GRel

```
inductive GRel (c : ℕ → ℕ) : FreeGroup (Gen c) → Prop
| H1a (s t : ℂ×) : GRel c (eqv (h1 s * h1 t) (h1 (s * t)))
| H1b (s t : ℂ×) : GRel c (eqv (h2 s * h2 t) (h2 (s * t)))
| H2 (s t : ℂ×) : GRel c (eqv (h1 s * h2 t) (h2 t * h1 s))
| Re1a (u v : ℂ) : GRel c (eqv (xm u * xm v) (xm (u + v)))
| Re1b (u v : ℂ) : GRel c (eqv (ym u * ym v) (ym (u + v)))
| Re2 (s t : ℂ×) : GRel c (eqv (ym (-(t : ℂ)) * xm (s : ℂ) * ym (t : ℂ))
  (xm (-(t : ℂ)-1) * ym (-(t : ℂ) ^ 2 * s) * xm ((t : ℂ)-1)))
| Re3 (s : ℂ×) : GRel c (eqv (wm s * wm1) (h1 (-s) * h2 (-(s-1))))
| Re4a (u : ℂ) : GRel c (eqv (conj' wm1 (xm u)) (ym (-u)))
| Re4b (u : ℂ) : GRel c (eqv (conj' wm1 (ym u)) (xm (-u)))
| Re5a (s : ℂ×) : GRel c (eqv (conj' wm1 (h1 s)) (h2 s))
| Re5b (s : ℂ×) : GRel c (eqv (conj' wm1 (h2 s)) (h1 s))
| Re6a (s : ℂ×) (u : ℂ) : GRel c (eqv (conj' (h1 s) (xm u)) (xm ((s : ℂ) * u)))
| Re6b (s : ℂ×) (u : ℂ) : GRel c (eqv (conj' (h2 s) (xm u)) (xm ((s : ℂ)-1 * u)))
| Re6c (s : ℂ×) (u : ℂ) : GRel c (eqv (conj' (h1 s) (ym u)) (ym ((s : ℂ)-1 * u)))
| Re6d (s : ℂ×) (u : ℂ) : GRel c (eqv (conj' (h2 s) (ym u)) (ym ((s : ℂ) * u)))
| Im1a (e : IdxE c) (u v : ℂ) : GRel c (eqv (xx e u * xx e v) (xx e (u + v)))
| Im1b (e : IdxE c) (u v : ℂ) : GRel c (eqv (yy e u * yy e v) (yy e (u + v)))
| Im2 (e : IdxE c) (s t : ℂ×) :
  GRel c (eqv (yy e (-(t : ℂ)) * xx e (s : ℂ) * yy e (t : ℂ))
    (xx e (-(t : ℂ)-1) / cc e.1.1 e.1.2.1 *
      yy e (-(cc e.1.1 e.1.2.1 * (t : ℂ) ^ 2 * (s : ℂ))) *

```

```

      xx e ((t : ℂ)-1 / cc e.1.1 e.1.2.1)))
| Im3 (e : IdxE c) (s : ℂ×) :
  GRel c (eqr (ww (c := c) e (s ^ ((e.1.1 + 1) * (e.1.2.1 - e.1.1))) * ww1 e)
    (h1 ((-s) ^ (e.1.2.1 - e.1.1)) * h2 ((-s) ^ (e.1.1 + 1))))))
| Im4a (e : IdxE c) (u : ℂ) :
  GRel c (eqr (conj' (ww1 e) (xx e u)) (yy e (-u / cc e.1.1 e.1.2.1)))
| Im4b (e : IdxE c) (u : ℂ) :
  GRel c (eqr (conj' (ww1 e) (yy e u)) (xx e (-(cc e.1.1 e.1.2.1) * u)))
| Im5a (e : IdxE c) (s : ℂ×) :
  GRel c (eqr (conj' (ww1 e) (h1 (s ^ (e.1.2.1 - e.1.1))) (h2 ((s ^ (e.1.1 + 1))-1)))
| Im5b (e : IdxE c) (s : ℂ×) :
  GRel c (eqr (conj' (ww1 e) (h2 (s ^ (e.1.1 + 1))) (h1 ((s ^ (e.1.2.1 - e.1.1))-1)))
| Im6a (e : IdxE c) (s : ℂ×) (u : ℂ) :
  GRel c (eqr (conj' (h1 s) (xx e u)) (xx e ((s : ℂ) ^ (e.1.1 + 1) * u)))
| Im6b (e : IdxE c) (s : ℂ×) (u : ℂ) :
  GRel c (eqr (conj' (h2 s) (xx e u)) (xx e ((s : ℂ) ^ (e.1.2.1 - e.1.1) * u)))
| Im6c (e : IdxE c) (s : ℂ×) (u : ℂ) :
  GRel c (eqr (conj' (h1 s) (yy e u)) (yy e (((s : ℂ) ^ (e.1.1 + 1))-1 * u)))
| Im6d (e : IdxE c) (s : ℂ×) (u : ℂ) :
  GRel c (eqr (conj' (h2 s) (yy e u)) (yy e (((s : ℂ) ^ (e.1.2.1 - e.1.1))-1 * u)))
| U1a (e : IdxE c) (u v : ℂ) (h : e.1.1 + 1 = e.1.2.1) :
  GRel c (comm' (xm u) (xx e v))
| U1b (e : IdxE c) (u v : ℂ) (h : e.1.1 + 1 = e.1.2.1) :
  GRel c (comm' (ym u) (yy e v))
| U1c (e : IdxE c) (u v : ℂ) (h : e.1.1 = 0) :
  GRel c (comm' (ym u) (xx e v))
| U1d (e : IdxE c) (u v : ℂ) (h : e.1.1 = 0) :
  GRel c (comm' (xm u) (yy e v))
| U2 (e f : IdxE c) (u v : ℂ)
  (h : e.1.2.1 ≠ f.1.2.1 ∨ e.1.2.2 ≠ f.1.2.2 ∨
    1 < |((e.1.1 : ℤ) - (f.1.1 : ℤ))|) :
  GRel c (comm' (xx e u) (yy f v))
| U3a (e f : IdxE c) (u v : ℂ)
  (he : e.1.1 = 1 ∧ e.1.2.1 = 2) (hf : f.1.1 = 0 ∧ f.1.2.1 = 2)
  (hk : e.1.2.2 = f.1.2.2) :
  GRel c (eqr (comm' (xx e u) (yy f v)) (xm (u * v)))
| U3b (e f : IdxE c) (u v : ℂ)
  (he : e.1.1 = 0 ∧ e.1.2.1 = 2) (hf : f.1.1 = 1 ∧ f.1.2.1 = 2)
  (hk : e.1.2.2 = f.1.2.2) :
  GRel c (eqr (comm' (xx e u) (yy f v)) (ym (u * v)))
| U4a (e : IdxE c) (u : ℂ) :
  GRel c (eqr (conj' wm1 (xx e u))
    (xx (flipE e) ((-1 : ℂ) ^ (e.1.2.1 - e.1.1 - 1) * u)))
| U4b (e : IdxE c) (u : ℂ) :
  GRel c (eqr (conj' wm1 (yy e u))
    (yy (flipE e) ((-1 : ℂ) ^ (e.1.2.1 - e.1.1 - 1) * u)))

```

Definition 2.19 (The group). $G_c = F(\mathcal{X}_c)/N_{\mathcal{R}_c}$, where $N_{\mathcal{R}_c}$ is the normal closure of $\mathcal{R}_c$. We write $\pi : F(\mathcal{X}_c) \rightarrow G_c$ for the quotient homomorphism.

Lean: Gm

```

| abbrev Gm (c : ℕ → ℕ) : Type := PresentedGroup {r : FreeGroup (Gen c) | GRel c r}

```


Target statement. Lean: target statement `gsimple_zero_generators_trivial`, fixed before the proof; solution is checked to have exactly this type

```

theorem gsimple_zero_generators_trivial (c : ℕ → ℕ) :
  ((PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm (c := c) 0) : Gm c) = 1) ∧
  ((PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (ym (c := c) 0) : Gm c) = 1) ∧
  (∀ e : IdxE c,
    (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xx e 0) : Gm c) = 1) ∧
  (∀ e : IdxE c,
    (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e 0) : Gm c) = 1)

```

The target statement asserts the following for every function $c: \mathbb{N} \rightarrow \mathbb{N}$:

- (a) $\pi(X_{-1}(0)) = 1$ in G_c ;
- (b) $\pi(Y_{-1}(0)) = 1$;
- (c) for every $e \in E_c$, $\pi(X_e(0)) = 1$;
- (d) for every $e \in E_c$, $\pi(Y_e(0)) = 1$.

This is [Lemma 1.1](#), with " $= 1$ in $G(\mathbf{m})$ " read as "the image under the quotient map is the identity". The function c is a free parameter, and no hypothesis on it is assumed; in particular $c(2) \geq 1$ is not needed here. Specializing c to the coefficients $c(j)$, $j \geq 0$, of J recovers the group $G(\mathbf{m})$ of the source, so the formal statement contains the source lemma.

3 Proof

3.1 Overview

The source proof sets $u = v = 0$ in (Re:1a) and cancels. The formal proof does the same, organized as follows.

- It isolates the cancellation step as an abstract fact ([Lemma 3.1](#)): any map $f: \mathbb{C} \rightarrow G$ into a group with $f(u + v) = f(u)f(v)$ satisfies $f(0) = 1$.
- It derives this additivity for $u \mapsto \pi(X_{-1}(u))$, $u \mapsto \pi(Y_{-1}(u))$, $u \mapsto \pi(X_e(u))$ and $u \mapsto \pi(Y_e(u))$. These are [Lemmas 3.2](#) to [3.5](#), which use (Re:1a), (Re:1b), (Im:1a) and (Im:1b) respectively.

The only deviations from the source are presentational. Parts (b)–(d), which the source leaves to the reader, are written out. The additivity is first established for all u, v and only afterwards specialized to $u = v = 0$. No other relation of $\mathcal{R}_c$ is used.

Lemma 3.1 (`add_zero_group_lemma`). *Let G be a group and let $f: \mathbb{C} \rightarrow G$ be any map such that $f(u + v) = f(u)f(v)$ for all $u, v \in \mathbb{C}$. Then $f(0) = 1$.*

Lean: `add_zero_group_lemma` — proved by AIProver, job 1, session 1

```

theorem add_zero_group_lemma {G : Type*} [Group G] (f : ℂ → G)
  (h : ∀ u v : ℂ, f (u + v) = f u * f v) : f 0 = 1 := by
  have h0 : f 0 = f 0 * f 0 := by
    calc
      f 0 = f (0 + 0) := by rw [zero_add]
      _ = f 0 * f 0 := h 0 0

```

```

calc
  f 0 = f 0 * 1 := by simp
  _ = f 0 * (f 0 * (f 0)-1) := by simp
  _ = (f 0 * f 0) * (f 0)-1 := by rw [mul_assoc]
  _ = f 0 * (f 0)-1 := by rw [← h0]
  _ = 1 := by simp

```

Proof. Taking $u = v = 0$ gives $f(0) = f(0 + 0) = f(0)f(0)$. Hence

$$f(0) = f(0)(f(0)f(0))^{-1} = (f(0)f(0))f(0)^{-1} = f(0)f(0)^{-1} = 1. \quad \square$$

Lemma 3.2 (mk_xm_add). *Let $c: \mathbb{N} \rightarrow \mathbb{N}$ and $u, v \in \mathbb{C}$. Then in G_c*

$$\pi(X_{-1}(u + v)) = \pi(X_{-1}(u)) \pi(X_{-1}(v)).$$

Lean: mk_xm_add — proved by AIProver, job 2, session 0

```

theorem mk_xm_add (c : ℕ → ℕ) (u v : ℂ) :
  (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm (u + v) : FreeGroup (Gen c)) : Gm c) =
    PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm u : FreeGroup (Gen c)) *
      PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm v : FreeGroup (Gen c)) := by
  have hrel := GRel.Re1a (c := c) u v
  have h2 : PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm u * xm v) *
    (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm (u + v)))-1 = 1 :=
    PresentedGroup.one_of_mem hrel
  have h1 : PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm u * xm v) =
    PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm u) *
      PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm v) := by
    simp
    rw [h1] at h2
    exact (eq_of_mul_inv_eq_one h2).symm

```

Proof. By (Re:1a), $r = X_{-1}(u)X_{-1}(v)X_{-1}(u+v)^{-1} \in \mathcal{R}_c$, so $\pi(r) = 1$. Since π is a homomorphism,

$$\pi(X_{-1}(u)) \pi(X_{-1}(v)) \pi(X_{-1}(u + v))^{-1} = 1.$$

Multiplying on the right by $\pi(X_{-1}(u + v))$ gives $\pi(X_{-1}(u))\pi(X_{-1}(v)) = \pi(X_{-1}(u + v))$. $\square$

Lemma 3.3 (mk_ym_add). *Let $c: \mathbb{N} \rightarrow \mathbb{N}$ and $u, v \in \mathbb{C}$. Then in G_c*

$$\pi(Y_{-1}(u + v)) = \pi(Y_{-1}(u)) \pi(Y_{-1}(v)).$$

Lean: mk_ym_add — proved by AIProver, job 2, session 0

```

theorem mk_ym_add (c : ℕ → ℕ) (u v : ℂ) :
  (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (ym (u + v) : FreeGroup (Gen c)) : Gm c) =
    PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (ym u : FreeGroup (Gen c)) *
      PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (ym v : FreeGroup (Gen c)) := by
  have hrel : GRel c (eqr (ym u * ym v) (ym (u + v))) := GRel.Re1b u v
  have h_eq : (PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) =
    (PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c) := by
    have h_one : (PresentedGroup.mk {r | GRel c r} (eqr (ym u * ym v) (ym (u + v))) : Gm c) = 1 :=

```

```

PresentedGroup.one_of_mem hrel
have h_mul_one : (PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) *
  ((PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c))-1 = 1 := by
  calc
    (PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) *
      ((PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c))-1 =
        (PresentedGroup.mk {r | GRel c r} ((ym u * ym v) * (ym (u + v))-1) : Gm c) := by
          simp
        _ = (PresentedGroup.mk {r | GRel c r} (eqr (ym u * ym v) (ym (u + v)))) : Gm c := rfl
        _ = 1 := h_one
  calc
    (PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) =
      ((PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) * 1) := by simp
    _ = ((PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) *
      (((PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c))-1 *
        (PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c))) := by simp
    _ = ((PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) *
      ((PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c))-1) *
      (PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c) := by rw [mul_assoc]
    _ = 1 * (PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c) := by rw [h_mul_one]
    _ = (PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c) := by simp
  calc
    (PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c)
      = (PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) := by rw [← h_eq]
    _ = (PresentedGroup.mk {r | GRel c r} (ym u) : Gm c) *
      (PresentedGroup.mk {r | GRel c r} (ym v) : Gm c) :=
      (PresentedGroup.mk {r | GRel c r}).map_mul (ym u) (ym v)

```

Proof. By (Re:1b), $Y_{-1}(u)Y_{-1}(v)Y_{-1}(u+v)^{-1} \in \mathcal{R}_c$, so its image under π is 1. That is, $\pi(Y_{-1}(u)Y_{-1}(v))\pi(Y_{-1}(u+v))^{-1} = 1$. Multiplying on the right by $\pi(Y_{-1}(u+v))$ gives $\pi(Y_{-1}(u)Y_{-1}(v)) = \pi(Y_{-1}(u+v))$. Finally, $\pi(Y_{-1}(u)Y_{-1}(v)) = \pi(Y_{-1}(u))\pi(Y_{-1}(v))$ because π is a homomorphism. $\square$

Lemma 3.4 (mk_xx_add). *Let $c : \mathbb{N} \rightarrow \mathbb{N}$, $e \in E_c$ and $u, v \in \mathbb{C}$. Then in G_c*

$$\pi(X_e(u+v)) = \pi(X_e(u))\pi(X_e(v)).$$

Lean: mk_xx_add — proved by AIProver, job 1, session 0

```

theorem mk_xx_add (c : ℕ → ℕ) (e : IdxE c) (u v : ℂ) :
  (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xx e (u + v)) : Gm c) =
    PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xx e u) *
    PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xx e v) := by
  let rels : Set (FreeGroup (Gen c)) := {r | GRel c r}
  have h := GRel.Im1a e u v
  have hmem : (xx e u * xx e v) * (xx e (u + v))-1 ∈ rels := h
  have h_eq : (PresentedGroup.mk rels) (xx e u * xx e v) = (PresentedGroup.mk rels) (xx e (u + v)) :=
    PresentedGroup.mk_eq_mk_of_mul_inv_mem hmem
  calc
    (PresentedGroup.mk rels) (xx e (u + v)) = (PresentedGroup.mk rels) (xx e u * xx e v) := by
      rw [h_eq]
    _ = (PresentedGroup.mk rels) (xx e u) * (PresentedGroup.mk rels) (xx e v) := by
      rw [(PresentedGroup.mk rels).map_mul]

```

Proof. By (Im:1a) for the index e , $(X_e(u)X_e(v))X_e(u+v)^{-1} \in \mathcal{R}_c$. If $ab^{-1} \in \mathcal{R}_c$, then $\pi(a) = \pi(b)$. Hence $\pi(X_e(u)X_e(v)) = \pi(X_e(u+v))$. Since π is a homomorphism, the left-hand side equals $\pi(X_e(u))\pi(X_e(v))$. $\square$

Lemma 3.5 (mk_yy_add). *Let $c: \mathbb{N} \rightarrow \mathbb{N}$, $e \in E_c$ and $u, v \in \mathbb{C}$. Then in G_c*

$$\pi(Y_e(u+v)) = \pi(Y_e(u))\pi(Y_e(v)).$$

Lean: mk_yy_add — proved by AIProver, job 1, session 0

```

theorem mk_yy_add (c : ℕ → ℕ) (e : IdxE c) (u v : ℂ) :
  (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e (u + v)) : Gm c) =
    PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e u) *
      PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e v) := by
  have h := GRel.Im1b e u v
  have h_eq : PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e u * yy e v) =
    PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e (u + v)) :=
    PresentedGroup.mk_eq_mk_of_mul_inv_mem h
  calc
    (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e (u + v)) : Gm c)
      = PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e u * yy e v) := by
      rw [← h_eq]
    _ = PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e u) *
      PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e v) := by
      rw [map_mul]

```

Proof. The argument is identical to Lemma 3.4, using the relator $Y_e(u)Y_e(v)Y_e(u+v)^{-1} \in \mathcal{R}_c$ from (Im:1b). $\square$

3.2 Proof of the theorem

Lean: solution — main proof from the captain's sketch

```

theorem solution (c : ℕ → ℕ) :
  ((PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm (c := c) 0) : Gm c) = 1) ∧
  ((PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (ym (c := c) 0) : Gm c) = 1) ∧
  (∀ e : IdxE c,
    (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xx e 0) : Gm c) = 1) ∧
  (∀ e : IdxE c,
    (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e 0) : Gm c) = 1) := by
  refine ⟨?, ?, ?, ?⟩
  · exact add_zero_group_lemma
    (fun u : ℂ => (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm u : FreeGroup (Gen c)) : Gm c))
    (mk_xm_add c)
  · exact add_zero_group_lemma
    (fun u : ℂ => (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (ym u : FreeGroup (Gen c)) : Gm c))
    (mk_ym_add c)
  · intro e
    exact add_zero_group_lemma
    (fun u : ℂ => (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xx e u) : Gm c))
    (mk_xx_add c e)
  · intro e
    exact add_zero_group_lemma
    (fun u : ℂ => (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e u) : Gm c))
    (mk_yy_add c e)

```

Fix $c: \mathbb{N} \rightarrow \mathbb{N}$. Each part is an application of [Lemma 3.1](#) with $G = G_c$:

- (a) Take $f(u) = \pi(X_{-1}(u))$. It is additive by [Lemma 3.2](#), so $\pi(X_{-1}(0)) = 1$.
- (b) Take $f(u) = \pi(Y_{-1}(u))$. It is additive by [Lemma 3.3](#), so $\pi(Y_{-1}(0)) = 1$.
- (c) Given $e \in E_c$, take $f(u) = \pi(X_e(u))$. It is additive by [Lemma 3.4](#), so $\pi(X_e(0)) = 1$.
- (d) Given $e \in E_c$, take $f(u) = \pi(Y_e(u))$. It is additive by [Lemma 3.5](#), so $\pi(Y_e(0)) = 1$.

This proves all four parts of the target statement. Because c is arbitrary, it proves [Lemma 1.1](#) in particular. $\square$

4 Independent review

A referee model (claude-opus-5-5), which took no part in the formalization or the proof, compared the verified Lean file with the source. Verdict: **FAITHFUL**.

Definitions. I compared each definition with the source.

- `IdxE` encodes E as triples (ℓ, j, k) of naturals with $1 \leq k \leq c(j)$ and $\ell < j$. Here $c: \mathbb{N} \rightarrow \mathbb{N}$ is a parameter, as the formalization note allows. The case $j = 0$ gives no elements because $\ell < 0$ is impossible, so the convention for $\mathbb{N}$ does not matter.
- `Gen` encodes $\mathcal{X}$: H_1, H_2 on $\mathbb{C}^\times$, X_{-1}, Y_{-1} on $\mathbb{C}$, and $X_{\ell, jk}, Y_{\ell, jk}$ on $E \times \mathbb{C}$.
- `cc` is $(-1)^{\ell+1} \binom{j-1}{\ell} (\ell+1)(j-\ell)$. The truncated subtractions are harmless because $\ell < j$ on E . The value is nonzero on E , so the Lean convention for division by zero never arises.
- `wm`, `wm1`, `ww` and `ww1` match `(Re:0)` and `(Im:0)`.
- The helper `comm'` is $aba^{-1}b^{-1}$, the source convention. The helper `eqr` encodes $a = b$ as ab^{-1} , and `conj'` is aba^{-1} .
- `GRel`: I checked each constructor against its tag. The `H`, `Re` and `Im` relations agree, including the exponents in `(Im:3)`, `(Im:5)` and `(Im:6)`; for example $s^{-(\ell+1)}$ is written as $(s^{\ell+1})^{-1}$.
- The `U` relations also agree.
 - `(U:1a,b)` use $\ell + 1 = j$, and `(U:1c,d)` use $\ell = 0$.
 - `(U:2)` uses the disjunction $j \neq p$, $k \neq q$, or $|\ell - m| > 1$.
 - `(U:3a,b)` use the correct indices $(1, 2, k)$ and $(0, 2, k)$.
 - `(U:4a,b)` use `flipE`, which sends ℓ to $j - 1 - \ell$, with sign $(-1)^{j-\ell-1}$.
- `Gm` is the presented group $F(\mathcal{X})/N_{\mathcal{R}}$.

I found no discrepancy.

Statement. The theorem is quantified over every $c : \mathbb{N} \rightarrow \mathbb{N}$ and has no further hypotheses. It asserts that the images in Gm of the following are all 1:

- $X_{-1}(0)$ and $Y_{-1}(0)$;
- $X_{\ell,jk}(0)$ for every $e \in E$;
- $Y_{\ell,jk}(0)$ for every $e \in E$.

These are exactly parts (a) through (d).

To recover the source statement, instantiate c with the coefficients of J restricted to $j \in \mathbb{N}$; these are natural numbers. The value $c(-1)$ never enters E .

The conclusion is not vacuous. Parts (a) and (b) are unconditional, and parts (c) and (d) range over a nonempty E for the true c . The relation set also has no degenerate encoding.

Proof.

- `add_zero_group_lemma` shows that if $f(u + v) = f(u)f(v)$ in a group, then $f(0) = 1$. It does this by deriving $f(0) = f(0)f(0)$ and cancelling, which is exactly the source argument with $u = v = 0$.
- `mk_xm_add`, `mk_ym_add`, `mk_xx_add` and `mk_yy_add` turn the relations (Re:1a), (Re:1b), (Im:1a) and (Im:1b), each a member of the relation set, into multiplicativity of the image in Gm . They use `PresentedGroup.one_of_mem` or `PresentedGroup.mk_eq_mk_of_mul_inv_mem`.
- `solution` combines these four lemmas with `add_zero_group_lemma`.

The proof follows the source proof, and its note covering parts (b) to (d). No auxiliary lemma is weaker or stronger than the step it represents.

Concerns. None

A Complete Lean file

The verified file, as checked by Lean.

```
import Mathlib
set_option autoImplicit false

def IdxE (c : ℕ → ℕ) : Type := {x : ℕ × ℕ × ℕ // 1 ≤ x.2.2 ∧ x.2.2 ≤ c x.2.1 ∧ x.1 < x.2.1}

inductive Gen (c : ℕ → ℕ) : Type
| H1 : ℂ× → Gen c
| H2 : ℂ× → Gen c
| Xm : ℂ → Gen c
| Ym : ℂ → Gen c
| X : IdxE c → ℂ → Gen c
| Y : IdxE c → ℂ → Gen c

noncomputable def cc (l j : ℕ) : ℂ :=
  (-1 : ℂ) ^ (1 + 1) * (Nat.choose (j - 1) 1 : ℂ) * ((1 + 1 : ℕ) : ℂ) * ((j - 1 : ℕ) : ℂ)

def flipE {c : ℕ → ℕ} (e : IdxE c) : IdxE c :=
  ⟨(e.1.2.1 - 1 - e.1.1, e.1.2.1, e.1.2.2), (e.2.1, e.2.2.1, by
```

```

have h := e.2.2.2
show e.1.2.1 - 1 - e.1.1 < e.1.2.1
omega))

section
variable {c : ℕ → ℕ}

def h1 (s : ℂ×) : FreeGroup (Gen c) := FreeGroup.of (Gen.H1 s)
def h2 (s : ℂ×) : FreeGroup (Gen c) := FreeGroup.of (Gen.H2 s)
def xm (u : ℂ) : FreeGroup (Gen c) := FreeGroup.of (Gen.Xm u)
def ym (u : ℂ) : FreeGroup (Gen c) := FreeGroup.of (Gen.Ym u)
def xx (e : IdxE c) (u : ℂ) : FreeGroup (Gen c) := FreeGroup.of (Gen.X e u)
def yy (e : IdxE c) (u : ℂ) : FreeGroup (Gen c) := FreeGroup.of (Gen.Y e u)

noncomputable def wm (s : ℂ×) : FreeGroup (Gen c) := xm (c := c) s * ym (-(s : ℂ)-1) * xm (c := c) s
noncomputable def wm1 : FreeGroup (Gen c) := wm (c := c) 1

noncomputable def ww (e : IdxE c) (s : ℂ×) : FreeGroup (Gen c) :=
  xx e s * yy e (-(s : ℂ)-1) / cc e.1.1 e.1.2.1 * xx e s
noncomputable def ww1 (e : IdxE c) : FreeGroup (Gen c) := ww e 1

def comm' (a b : FreeGroup (Gen c)) : FreeGroup (Gen c) := a * b * a-1 * b-1
def eqr (a b : FreeGroup (Gen c)) : FreeGroup (Gen c) := a * b-1
def conj' (a b : FreeGroup (Gen c)) : FreeGroup (Gen c) := a * b * a-1
end

inductive GRel (c : ℕ → ℕ) : FreeGroup (Gen c) → Prop
| H1a (s t : ℂ×) : GRel c (eqr (h1 s * h1 t) (h1 (s * t)))
| H1b (s t : ℂ×) : GRel c (eqr (h2 s * h2 t) (h2 (s * t)))
| H2 (s t : ℂ×) : GRel c (eqr (h1 s * h2 t) (h2 t * h1 s))
| Re1a (u v : ℂ) : GRel c (eqr (xm u * xm v) (xm (u + v)))
| Re1b (u v : ℂ) : GRel c (eqr (ym u * ym v) (ym (u + v)))
| Re2 (s t : ℂ×) : GRel c (eqr (ym (-(t : ℂ)) * xm (s : ℂ) * ym (t : ℂ))
  (xm (-(t : ℂ)-1) * ym (-(t : ℂ) ^ 2 * s) * xm ((t : ℂ)-1)))
| Re3 (s : ℂ×) : GRel c (eqr (wm s * wm1) (h1 (-s) * h2 (-(s-1))))
| Re4a (u : ℂ) : GRel c (eqr (conj' wm1 (xm u)) (ym (-u)))
| Re4b (u : ℂ) : GRel c (eqr (conj' wm1 (ym u)) (xm (-u)))
| Re5a (s : ℂ×) : GRel c (eqr (conj' wm1 (h1 s)) (h2 s))
| Re5b (s : ℂ×) : GRel c (eqr (conj' wm1 (h2 s)) (h1 s))
| Re6a (s : ℂ×) (u : ℂ) : GRel c (eqr (conj' (h1 s) (xm u)) (xm ((s : ℂ) * u)))
| Re6b (s : ℂ×) (u : ℂ) : GRel c (eqr (conj' (h2 s) (xm u)) (xm ((s : ℂ)-1 * u)))
| Re6c (s : ℂ×) (u : ℂ) : GRel c (eqr (conj' (h1 s) (ym u)) (ym ((s : ℂ)-1 * u)))
| Re6d (s : ℂ×) (u : ℂ) : GRel c (eqr (conj' (h2 s) (ym u)) (ym ((s : ℂ) * u)))
| Im1a (e : IdxE c) (u v : ℂ) : GRel c (eqr (xx e u * xx e v) (xx e (u + v)))
| Im1b (e : IdxE c) (u v : ℂ) : GRel c (eqr (yy e u * yy e v) (yy e (u + v)))
| Im2 (e : IdxE c) (s t : ℂ×) :
  GRel c (eqr (yy e (-(t : ℂ)) * xx e (s : ℂ) * yy e (t : ℂ))
    (xx e (-(t : ℂ)-1) / cc e.1.1 e.1.2.1 *
      yy e (-(cc e.1.1 e.1.2.1 * (t : ℂ) ^ 2 * (s : ℂ))) *
      xx e ((t : ℂ)-1 / cc e.1.1 e.1.2.1)))
| Im3 (e : IdxE c) (s : ℂ×) :
  GRel c (eqr (ww (c := c) e (s ^ ((e.1.1 + 1) * (e.1.2.1 - e.1.1))) * ww1 e)
    (h1 ((-s) ^ (e.1.2.1 - e.1.1)) * h2 ((-s) ^ (e.1.1 + 1))))
| Im4a (e : IdxE c) (u : ℂ) :
  GRel c (eqr (conj' (ww1 e) (xx e u)) (yy e (-u / cc e.1.1 e.1.2.1)))
| Im4b (e : IdxE c) (u : ℂ) :
  GRel c (eqr (conj' (ww1 e) (yy e u)) (xx e (-(cc e.1.1 e.1.2.1 * u)))
| Im5a (e : IdxE c) (s : ℂ×) :
  GRel c (eqr (conj' (ww1 e) (h1 (s ^ (e.1.2.1 - e.1.1)))) (h2 ((s ^ (e.1.1 + 1))-1)))

```

```

| Im5b (e : IdxE c) (s :  $\mathbb{C}^{\times}$ ) :
  GRel c (eqr (conj' (ww1 e) (h2 (s ^ (e.1.1 + 1)))) (h1 ((s ^ (e.1.2.1 - e.1.1))-1)))
| Im6a (e : IdxE c) (s :  $\mathbb{C}^{\times}$ ) (u :  $\mathbb{C}$ ) :
  GRel c (eqr (conj' (h1 s) (xx e u)) (xx e ((s :  $\mathbb{C}$ ) ^ (e.1.1 + 1) * u)))
| Im6b (e : IdxE c) (s :  $\mathbb{C}^{\times}$ ) (u :  $\mathbb{C}$ ) :
  GRel c (eqr (conj' (h2 s) (xx e u)) (xx e ((s :  $\mathbb{C}$ ) ^ (e.1.2.1 - e.1.1) * u)))
| Im6c (e : IdxE c) (s :  $\mathbb{C}^{\times}$ ) (u :  $\mathbb{C}$ ) :
  GRel c (eqr (conj' (h1 s) (yy e u)) (yy e ((s :  $\mathbb{C}$ ) ^ (e.1.1 + 1))-1 * u)))
| Im6d (e : IdxE c) (s :  $\mathbb{C}^{\times}$ ) (u :  $\mathbb{C}$ ) :
  GRel c (eqr (conj' (h2 s) (yy e u)) (yy e ((s :  $\mathbb{C}$ ) ^ (e.1.2.1 - e.1.1))-1 * u)))
| U1a (e : IdxE c) (u v :  $\mathbb{C}$ ) (h : e.1.1 + 1 = e.1.2.1) :
  GRel c (comm' (xm u) (xx e v))
| U1b (e : IdxE c) (u v :  $\mathbb{C}$ ) (h : e.1.1 + 1 = e.1.2.1) :
  GRel c (comm' (ym u) (yy e v))
| U1c (e : IdxE c) (u v :  $\mathbb{C}$ ) (h : e.1.1 = 0) :
  GRel c (comm' (ym u) (xx e v))
| U1d (e : IdxE c) (u v :  $\mathbb{C}$ ) (h : e.1.1 = 0) :
  GRel c (comm' (xm u) (yy e v))
| U2 (e f : IdxE c) (u v :  $\mathbb{C}$ )
  (h : e.1.2.1  $\neq$  f.1.2.1  $\vee$  e.1.2.2  $\neq$  f.1.2.2  $\vee$ 
    1 < |((e.1.1 :  $\mathbb{Z}$ ) - (f.1.1 :  $\mathbb{Z}$ ))|) :
  GRel c (comm' (xx e u) (yy f v))
| U3a (e f : IdxE c) (u v :  $\mathbb{C}$ )
  (he : e.1.1 = 1  $\wedge$  e.1.2.1 = 2) (hf : f.1.1 = 0  $\wedge$  f.1.2.1 = 2)
  (hk : e.1.2.2 = f.1.2.2) :
  GRel c (eqr (comm' (xx e u) (yy f v)) (xm (u * v)))
| U3b (e f : IdxE c) (u v :  $\mathbb{C}$ )
  (he : e.1.1 = 0  $\wedge$  e.1.2.1 = 2) (hf : f.1.1 = 1  $\wedge$  f.1.2.1 = 2)
  (hk : e.1.2.2 = f.1.2.2) :
  GRel c (eqr (comm' (xx e u) (yy f v)) (ym (u * v)))
| U4a (e : IdxE c) (u :  $\mathbb{C}$ ) :
  GRel c (eqr (conj' wm1 (xx e u))
    (xx (flipE e) ((-1 :  $\mathbb{C}$ ) ^ (e.1.2.1 - e.1.1 - 1) * u)))
| U4b (e : IdxE c) (u :  $\mathbb{C}$ ) :
  GRel c (eqr (conj' wm1 (yy e u))
    (yy (flipE e) ((-1 :  $\mathbb{C}$ ) ^ (e.1.2.1 - e.1.1 - 1) * u)))

```

abbrev Gm (c : $\mathbb{N} \rightarrow \mathbb{N}$) : **Type** := PresentedGroup {r : FreeGroup (Gen c) | GRel c r}

theorem add_zero_group_lemma {G : **Type** *} [Group G] (f : $\mathbb{C} \rightarrow G$)

(h : $\forall u v : \mathbb{C}, f(u + v) = f u * f v$) : f 0 = 1 := **by**

have h0 : f 0 = f 0 * f 0 := **by**

calc

f 0 = f (0 + 0) := **by** rw [zero_add]

_ = f 0 * f 0 := h 0 0

calc

f 0 = f 0 * 1 := **by** simp

_ = f 0 * (f 0 * (f 0)⁻¹) := **by** simp

_ = (f 0 * f 0) * (f 0)⁻¹ := **by** rw [mul_assoc]

_ = f 0 * (f 0)⁻¹ := **by** rw [← h0]

_ = 1 := **by** simp

theorem mk_xm_add (c : $\mathbb{N} \rightarrow \mathbb{N}$) (u v : $\mathbb{C}$) :

(PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm (u + v) : FreeGroup (Gen c)) : Gm c) =

PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm u : FreeGroup (Gen c)) *

PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm v : FreeGroup (Gen c)) := **by**

have hrel := GRel.Re1a (c := c) u v


```

have h2 : PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm u * xm v) *
  (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm (u + v)))-1 = 1 :=
  PresentedGroup.one_of_mem hrel
have h1 : PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm u * xm v) =
  PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm u) *
  PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm v) := by
  simp
rw [h1] at h2
exact (eq_of_mul_inv_eq_one h2).symm

theorem mk_ym_add (c : ℕ → ℕ) (u v : ℂ) :
  (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (ym (u + v) : FreeGroup (Gen c)) : Gm c) =
  PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (ym u : FreeGroup (Gen c)) *
  PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (ym v : FreeGroup (Gen c)) := by

have hrel : GRel c (eqr (ym u * ym v) (ym (u + v))) := GRel.Re1b u v
have h_eq : (PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) =
  (PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c) := by
have h_one : (PresentedGroup.mk {r | GRel c r} (eqr (ym u * ym v) (ym (u + v))) : Gm c) = 1 :=
  PresentedGroup.one_of_mem hrel
have h_mul_one : (PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) *
  ((PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c))-1 = 1 := by
  calc
    (PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) *
      ((PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c))-1 =
      (PresentedGroup.mk {r | GRel c r} ((ym u * ym v) * (ym (u + v))-1) : Gm c) := by
        simp
      _ = (PresentedGroup.mk {r | GRel c r} (eqr (ym u * ym v) (ym (u + v))) : Gm c) := rfl
      _ = 1 := h_one
  calc
    (PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) =
      ((PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) * 1) := by simp
    _ = ((PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) *
      (((PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c))-1 *
      (PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c))) := by simp
    _ = ((PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) *
      ((PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c))-1 *
      (PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c)) := by rw [mul_assoc]
    _ = 1 * (PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c) := by rw [h_mul_one]
    _ = (PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c) := by simp
  calc
    (PresentedGroup.mk {r | GRel c r} (ym (u + v)) : Gm c)
      = (PresentedGroup.mk {r | GRel c r} (ym u * ym v) : Gm c) := by rw [← h_eq]
    _ = (PresentedGroup.mk {r | GRel c r} (ym u) : Gm c) *
      (PresentedGroup.mk {r | GRel c r} (ym v) : Gm c) :=
      (PresentedGroup.mk {r | GRel c r}).map_mul (ym u) (ym v)

theorem mk_xx_add (c : ℕ → ℕ) (e : IdxE c) (u v : ℂ) :
  (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xx e (u + v)) : Gm c) =
  PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xx e u) *
  PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xx e v) := by
let rels : Set (FreeGroup (Gen c)) := {r | GRel c r}
have h := GRel.Im1a e u v
have hmem : (xx e u * xx e v) * (xx e (u + v))-1 ∈ rels := h
have h_eq : (PresentedGroup.mk rels) (xx e u * xx e v) = (PresentedGroup.mk rels) (xx e (u + v)) :=
  PresentedGroup.mk_eq_mk_of_mul_inv_mem hmem
calc
  (PresentedGroup.mk rels) (xx e (u + v)) = (PresentedGroup.mk rels) (xx e u * xx e v) := by
    rw [h_eq]

```

```

_ = (PresentedGroup.mk rels) (xx e u) * (PresentedGroup.mk rels) (xx e v) := by
  rw [(PresentedGroup.mk rels).map_mul]

theorem mk_yy_add (c :  $\mathbb{N} \rightarrow \mathbb{N}$ ) (e : IdxE c) (u v :  $\mathbb{C}$ ) :
  (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e (u + v)) : Gm c) =
    PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e u) *
    PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e v) := by
have h := GRel.Im1b e u v
have h_eq : PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e u * yy e v) =
  PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e (u + v)) :=
    PresentedGroup.mk_eq_mk_of_mul_inv_mem h
calc
  (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e (u + v)) : Gm c)
    = PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e u * yy e v) := by
    rw [← h_eq]
  _ = PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e u) *
    PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e v) := by
    rw [map_mul]

theorem solution (c :  $\mathbb{N} \rightarrow \mathbb{N}$ ) :
  ((PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm (c := c) 0) : Gm c) = 1) ∧
  ((PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (ym (c := c) 0) : Gm c) = 1) ∧
  (∀ e : IdxE c,
    (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xx e 0) : Gm c) = 1) ∧
  (∀ e : IdxE c,
    (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e 0) : Gm c) = 1) := by
refine ⟨?, ?, ?, ?⟩
· exact add_zero_group_lemma
  (fun u :  $\mathbb{C} \Rightarrow$  (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xm u : FreeGroup (Gen c)) : Gm c))
  (mk_xm_add c)
· exact add_zero_group_lemma
  (fun u :  $\mathbb{C} \Rightarrow$  (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (ym u : FreeGroup (Gen c)) : Gm c))
  (mk_ym_add c)
· intro e
  exact add_zero_group_lemma
  (fun u :  $\mathbb{C} \Rightarrow$  (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (xx e u) : Gm c))
  (mk_xx_add c e)
· intro e
  exact add_zero_group_lemma
  (fun u :  $\mathbb{C} \Rightarrow$  (PresentedGroup.mk {r : FreeGroup (Gen c) | GRel c r} (yy e u) : Gm c))
  (mk_yy_add c e)

```